Sealed Classes in C#

1️⃣ What is a Sealed Class?

A **sealed class** is a class that **cannot be inherited** by another class.

Once a class is declared as sealed, **no other class can derive from it**.

Definition

A **sealed class prevents inheritance** and stops other classes from extending it.

2️⃣ Syntax of Sealed Class

```
sealed class ClassName
{
   // members
}
```

Example:

```
sealed class Calculator
{
   public int Add(int a, int b)
   {
      return a + b;
   }
}
```

Now if someone tries to inherit:

```
class AdvancedCalculator : Calculator
{
}
```

❌ This will produce a **compile-time error**.

Error:

Cannot derive from sealed type 'Calculator'

3️⃣ Example Program

Sealed Class

```
using System;


sealed class Logger
{
    public void Log(string message)
    {
        Console.WriteLine("Log: " + message);
    }
}
```

Main Program

```
class Program
{
    static void Main()
    {
        Logger log = new Logger();
        log.Log("Application Started");
    }
}
```

Output

Log: Application Started

But **no class can inherit Logger**.

4️⃣ Why Do We Use Sealed Classes?

1️⃣ Prevent Further Inheritance

Sometimes a class is **complete and should not be modified**.

Example:

Security related classes
Utility classes

2️⃣ Protect Implementation

It prevents developers from **changing or extending critical logic**.

Example:

Payment processing
Authentication classes

5️⃣ Real World Example

Example: **Bank Account System**

```
sealed class ATMTransaction
{
    public void ProcessTransaction()
    {
        Console.WriteLine("Processing ATM Transaction");
    }
}
```

If someone tries:

```
class OnlineTransaction : ATMTransaction
```

```
{

}
```

❌ Not allowed.

Because **ATMTransaction logic should not be extended**.

6️⃣ Sealed Method

A **method can also be sealed**, but only in **derived classes when overriding**.

Meaning:
A method **cannot be overridden further**.

Example

```
class Parent

{

   public virtual void Show()

   {

      Console.WriteLine("Parent Method");

   }

}


class Child : Parent

{

   public sealed override void Show()

   {

      Console.WriteLine("Child Method");

   }

}
```

Now if another class tries:

```
class GrandChild : Child
{
    public override void Show()  // ❌ Error
    {
        Console.WriteLine("Grandchild Method");
    }
}
```

❌ Error because method is **sealed**.

7️⃣ Important Rules of Sealed Class

Rule	Description
Cannot inherit sealed class	Yes
Can create objects	Yes
Can contain methods	Yes
Can contain properties	Yes
Can implement interfaces	Yes

Example:

```
sealed class PaymentService : IPayment
{
}
```

8️⃣ When Should You Use Sealed Class?

Use sealed class when:

✓ Class should not be extended
✓ Security is important
✓ Business logic must not be modified
✓ Performance optimization needed

9️⃣ Example from .NET Framework

Many .NET classes are sealed.

Example:

- String
- Math
- Console

Example:

```
string name = "Hello";
```

You **cannot inherit from String class**.

Quick Comparison

Feature	Normal Class	Sealed Class
Can be inherited	Yes	❌ No
Can create objects	Yes	Yes
Used for extension	Yes	No

✅ Simple Example

```
sealed class Car
{
    public void Start()
```

```
    {
        Console.WriteLine("Car Started");
    }
}
```

❌ Not allowed:

```
class SportsCar : Car
{
}
```